CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. Examination (EE)

May 2013

EE403 Power Electronics & Drives II

Date: 08.05.2013, Wednesday **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Draw and explain two stage sequence controlled AC voltage controller with R load. And find out equation for rms output voltage and input pf. [07]

Q - 2 (a) Draw and explain Stator frequency control for three-phase Induction motor with its characteristics. [07]

(b) Draw and explain Static Rotor-resistance control for three-phase SRIM. [07]

OR

Q – 2 (a) Draw and explain principle of Integral Cycle control. And also find out equation for rms output voltage and input pf. [07]

(b) Explain single-phase to single-phase step-down bridge type cycloconverter with R load. Also draw all gate pulses for all switching devices [07]

Q – 3 (a) A single-phase half-wave ac voltage controller feeds a load of R = 20 Ω with an input voltage of 230 V, 50 Hz. Firing angle of thyristor is 45^0. Determine (i) rms value of output voltage and (ii) average input current. [05]

(b) Explain single-phase step-up mid-point Cycloconverter & draw its waveforms. [05]

(c) What do you mean by Cycloconverter? List out applications of it. [04]

OR

Q - 3 (a) A 3-phase, 400 V, 15kW, 1440 rpm, 50 Hz, star-connected induction motor has rotor leakage impedance of 0.4+ j1.6 Ω. Stator leakage impedance and rotational losses are assumed negligible. If this motor is energized form 120 Hz, 400 V, 3-phase source, then calculate (i) The motor speed at rated load (ii) The slip at which max. torque occurs and (iii) The max. Torque. [10]

(b) A single-phase bridge-type cycloconverter has input voltage of 230 V, 50 Hz and load of R = 10 Ω. Output frequency is on-third of input frequency. For a firing angle delay of 30^0, calculate (i) rms output voltage (ii) rms current of each converter and (c) input power factor. [04]

SECTION – II

Q - 4 Draw and explain the harmonic reduction in the inverter output voltage by transformer Connections. [07]

Q - 5 (a) Describe 3-phase 120^0 mode Voltage source Inverter. Draw the waveforms of all gate pulses, line to line voltages & phase voltages. [10]

(b) Explain Push-pull converter with neat diagram. [04]

OR

Q - 5 (a) A sigle-phase ac switch of Fig.1 is used in between a 230 V, 50 Hz source and a load of 2 kW at pf of 0.8 lagging. Determine the voltage and current rating of (i) the thyristor and (ii) diodes of the bridge. Take a factor of safety of two. [07]

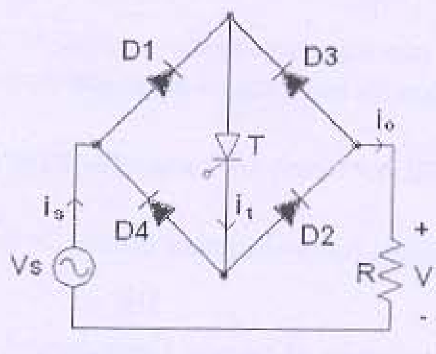

Fig.1

(b) Draw and Explain 12 pulse converter with neat diagram and waveforms. [07]

Q - 6 (a) A Three-phase bridge inverter delivers power to a resistive load from a 450 V dc source. For a star-connected load of 10 Ω per phase, determine for 180^0 mode (also draw necessary waveforms) (i) rms value of output line voltage V_L (ii) rms value of output phase voltage V_P (iii) rms value of load current (iv) rms value of Thyristor current and (v) Load Power [10]

(b) Draw and Explain Online UPS with its advantages and Applications. [04]

OR

Q - 6 (a) Draw and explain AC and DC Solid State Relays. [06]

(b) What do you mean by Static Switches? Why they are replacing mechanical and electromechanical switches? [03]

(c) Draw and Explain 6 pulse converter with neat diagram and waveforms. [05]

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